MCC Entrance Test, August 2026 — Day 1

Level T4

Test Information. Level T4 — 180 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra, Combinatorics (×2), Geometry (×2), Number Theory.
Rules: show your work (full, justified arguments earn full credit at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts). Positive integers a and b satisfy $a^5 = b^4$, and positive integers c and d satisfy $c^3 = d^2$. Given that $c - a = 19$, find $d - b$.

Problem 2 (10 pts). Find the remainder when 2^{1990} is divided by 1990.

Problem 3 (15 pts). For each integer $0 \leq n \leq 11$, there are exactly three identical weights weighing 2^n grams. Two piles of weights are considered different if they contain different numbers of weights of some size.

(a) Suppose only the weights weighing 1, 2, and 4 grams (three of each) are available. In how many ways can a pile weighing 6 grams be formed? [4 pts]

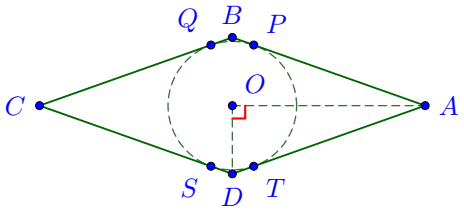
(b) Using all twelve weights $2^0, 2^1, \ldots, 2^{11}$ (three of each), in how many ways can a pile weighing 2021 grams be formed? [11 pts]

Problem 4 (15 pts).

A circle has centre O and radius 1. Quadrilateral $ABCD$ has all four sides tangent to the circle, at points P (on AB), Q (on BC), S (on CD), and T (on DA), as shown. Also, $\angle AOB = \angle BOC = \angle COD = \angle DOA$. Given that $AO = 3$,

(a) Find the length of AT. [6 pts]

(b) Find the length of DS. [9 pts]

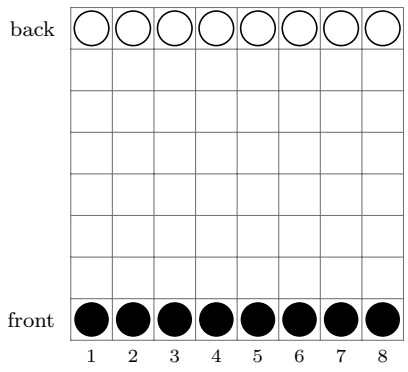


Problem 5 (25 pts). Let $ABCD$ be a convex quadrilateral with $\angle DAC = 36°$, $\angle BDC = 36°$, $\angle CBD = 18°$, and $\angle BAC = 72°$. The diagonals AC and BD intersect at point P.

(a) Prove that $AB = AC = AD$. [12 pts]

(b) Find $\angle ACB$ and $\angle ADC$. [3 pts]

(c) Find $\angle APD$. [10 pts]



Problem 6 (25 pts). On an 8×8 grid, eight Black tokens stand on the front rank and eight White tokens stand on the back rank. Each token moves like a queen in chess – any number of squares horizontally, vertically, or diagonally, so long as no other token blocks its path – and Black and White tokens move alternately, one token at a time.

(a) Show that at least 16 moves in total are required to interchange the two ranks. [3 pts]

(b) Show that at least 22 moves in total are required. [7 pts]

(c) Determine the minimum number of moves required with a sequence of moves achieving it. [15 pts]